Anikait Kaushik

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LinkedIn | GitHub | Portfolio | Projects & Credentials

Objective

To build intelligent robotic systems that augment human capability, address complex real-world challenges, and translate engineering innovation into practical solutions with meaningful societal impact.

Education

Thapar Institute of Engineering and Technology, Patiala (Punjab) **Aug 2023 – May 2027**
B.E. in Mechatronics Engineering; ***CGPA: 9.29/10.0*** *through 6th Semester*

GMSSS MHC, Sector 13, Chandigarh **Mar 2022 – Apr 2023**
Class XII; PCME; ***85.4%***

Jainendra Public School, Panchkula **Mar 2020 – Apr 2021**
Class X; ***96.4%***

Achievements

- **Champion** – International Rover Challenge 2026, SPROS, MIT Manipal; 1st of 33 finalist teams .
- **1st Runner-Up** – International Space Drone Challenge 2026, SPROS, MIT Manipal; 2nd of 20 finalist teams .
- **1st Runner-Up** – International Rover Challenge 2025, SPROS, Goa; 2nd of 31 finalist teams .
- **Department Topper – 10.0/10.0 CGPA** in 4th Semester, Department of Mechanical Engineering, TIET, Patiala.
- **Certificate of Achievement** – Orangewood Labs, for outstanding contribution and dedication during the internship.
- **Merit Scholar (1st–4th Year)** – TIET; Scholarships: Rs. 75,450 / Rs. 1,41,000 / Rs. 1,79,400 / Rs. 1,53,300.

Patents & Publications

- **Published Patent Application (2025):** *An Apparatus, System and Method for Robotic Soil Analysis and Collection*; Named Inventor, Application No. 202511062827.
- **IEEE Transactions on Robotics – Under Review:** *Experimental Validation of a Unified Euler–Lagrange Framework Using a Three-Link Leg Model for Human Gait Analysis; (Impact Factor - 11.1).*
- **Journal of Biomechanics – Manuscript in Final Preparation:** *Load-Dependent Changes in Lower-Limb Joint Kinetics and Hip–Knee Coordination During Walking; (Impact Factor - 2.9).*
- **Chapter Accepted for Publication – Bentham Science:** *Urban Air Quality Management through AI-Driven Decision Support*, in *Smart Approaches to Air Remediation: AI Applications in Air Pollution Control and Monitoring.*
- **Chapter Accepted for Publication – Bentham Science:** *Industrial and Municipal Case Studies on AI-Enabled Water Cleanup*, in *Intelligent Systems for Water Remediation: Transforming Wastewater and Water Quality Management.*
- **IPRoMM 2026 – Under Review:** *Design and Development of a Multi-Axis Robotic Subsystem for In-Situ Soil Sampling and Analysis.*

Leadership

Mechatronics and Robotics Society (MARS), TIET **Aug 2025 – Aug 2026**
General Secretary & Team Captain – Team MARS *Patiala, Punjab*

- Led and mentored 150+ students across robotics, mechatronics, mechanical design, and research; oversaw a Rs.8,00,000 sanctioned annual budget and Rs.16,00,000+ in competition-related expenditure.
- Directed multidisciplinary teams for the International Rover Challenge and International Space Drone Challenge; initiated DD Robocon at MARS and led MARVIN Gen 2 and Reconfigurable Robot Gen 1 .

Projects

CLIMB CART | Capstone Engineering Project *Mentors: Prof. Ashish Singla, Prof. Dheeraj Gupta*

- Engineered a stair-climbing load carrier for narrow, variable Indian staircases, translating field-surveyed constraints on geometry, payload, terrain, portability, and maintainability into the mechanical design through CAD, FEA, prototyping, and experimental validation.

Ganga | 4th Gen. Mars Rover | IRC'26 *Mentors: Prof. Ashish Singla, Dr. Sachin Kansal*

- Served as Team Captain, leading overall mechanical development and overseeing the 5-DOF manipulator, ABEx soil-sampling subsystem, and PU wheel integration from Wheeleez Australia; achieved 150 mm soil penetration, 12 g sample recovery, and 21/30 successful trials in IRC 2026.

Alaknanda | 3rd Gen. Mars Rover | IRC'25 *Mentors: Prof. Ashish Singla, Dr. Sachin Kansal*

- Designed and fabricated the complete AbEx soil-collection and testing subsystem, integrating soil extraction, sample handling, chemical testing, and sensor interfaces; validated components using SolidWorks, FEA, 3D printing, and CNC machining, earning highest AbEx points and judge recognition as the best-designed soil testing system at IRC 2025.

MATLAB–ROS Integration | LoCoBot *Mentors: Prof. Ashish Singla, Dr. Pranav Lad (MathWorks)*

- Developing an MATLAB–ROS framework using MATLAB Copilot and a Simscape-based digital twin of the Trossen Robotics LoCoBot, integrating robot kinematics, dynamics, sensing, and control interfaces to establish a Sim2Real workflow for simulation-driven development and physical validation, in collaboration with MathWorks.

Experience

The University of Queensland (UQ) – TIET-UQ Centre of Excellence **Jun 2025 – Present**
Research Intern – R3: Exoskeleton-enabled Rehabilitation

Mentors: Prof. Ashish Singla, Dr. Sachin Kansal, Dr. Alejandro Melendez-Calderon (UQ)

- Contributing to **R3 (Realize, Revive & Recover) – AI-Enabled Data-Driven Solutions for Exoskeleton-enabled Rehabilitation**, developing a custom wearable lower-limb exoskeleton across robotics, HRC, and biomechanics.
- Developed the experimental workflow in the Gait Laboratory using Qualisys motion capture, AMTI force plates, and Noraxon EMG for synchronized analysis of lower-limb kinematics, gait cycles, GRF, COP, and muscle activity during exoskeleton-assisted gait experiments.

Orangewood Labs **Jun 2026**
Robotics Systems Intern

Mentors: Manan Gupta, Rohit Kumar (Orangewood Labs)

- Completed the Robotics & Vision Systems Training & Internship Program by TIET and Orangewood Labs; programmed and validated pick-and-place operations on an Orangewood arm using the OWL GUI, explored RoboGPT for AI-assisted programming and interaction, and received a Certificate of Achievement.

Experiential Learning Centre (ELC), TIET **Jun 2024 – Jul 2024**
Summer Intern – MARVIN Autonomous Cleaning Robot

Mentors: Prof. Ashish Singla, Prof. R.K. Duvedi

- Designed and developed the mechanical system of MARVIN, including chassis, component mounting, and vacuum-cleaning mechanism prototype; integrated obstacle-avoidance hardware and Arduino control, and carried out SolidWorks CAD, 3D printing, sheet-metal fabrication, assembly, and testing; recognized among the Top 10 ELC Summer Internship Projects.

Technical Skills

Mechanical Design & CAD: SolidWorks, PTC Creo Parametric, AutoCAD, Mechanical Design, Prototyping
Simulation, Analysis & Robotics: MATLAB, Simulink, Simscape, ANSYS Workbench, FEA, Forward/Inverse Kinematics, Robot Dynamics, ROS 2 (Fundamentals), RoboGPT
Human Biomechanics & Gait Analysis: Sensor Integration, CubeMars Actuators, Qualisys Track Manager (QTM), Visual3D, AMTI Force Plates, Noraxon EMG
Manufacturing & Fabrication: Additive Manufacturing, 3D Printing, CNC Machining, Sheet-Metal Fabrication, PCB Fabrication
Programming & Software: Python, C, C++ *(basic working knowledge)*, Arduino IDE, Visual Studio Code, Git, GitHub

Certifications & Professional Development

- **Active Member – The Robotics Society (TRS)** – Membership Number: S254791.
- **TRS–DSAI International Workshop on Intelligent Systems: The Future of Robotics and AI – Organizing Committee**, TIET, Oct. 30–31, 2026.
- **Robotics & AI Training Program** – Orangewood Labs & TIET, 2 weeks, 2026.
- **An Easy Introduction to FEA with Applications using CAE** – 16-hour course, Udemy, 2026.
- **Dynamics of Robotic Systems** – Advances in Robotics (AIR), TRS, IIT Jodhpur, June 2025.
- **Practicing Robotics: Reconfigurable Robotics (ROS)** – TRS, IIT Ropar, March 2025.
- **MATLAB Onramp** – MathWorks Training Services, 2025.
- Hands-on exposure to **robotic system installation, commissioning, and verification** during deployment of Unitree Go2 Pro EDU, Addverb Synchro 5 Cobot, Alpha Alice Service Robot, Orangewood arm, and a exoskeleton.